ALGEBRAIZATION OF TRUNCATED DERIVED LOCAL COHOMOLOGY UNDER DEPTH AND DIMENSION BOUNDS

THE STACKS PROJECT AUTHORS

Editorial note. All five hypothesis groups, including the three-way alternative in condition5, remain verbatim. The source future-reference footnote is not used as a proof dependency.

Editorial note. Public310 is an annihilator/depth equivalence. This task constructs a support ideal making a derived-completion comparison an isomorphism in a specified range; it is not that annihilator theorem again.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Noetherian local cohomology, dualizing-complex dimension functions, derivedI-completion, depth and support, local duality and annihilator criteria. The two local kill-completion/kill-colimit proofs are retained.

Source reference. Original lemma 0955: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11573–11573. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0A6W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12859–12859. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BJC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14520–14520. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 091V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11458–11458. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0956: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11574–11574. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0EFE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18059–18059. Stable Stacks tag for an expressly

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allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DWY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17413–17413. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0ECM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17962–17962. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DWV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17410–17410. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AVZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13702–13702. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0EF1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18046–18046. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 0EFC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18057–18057. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 0EFD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18058–18058. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7062–7062. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DXB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17426–17426. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DXF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17430–17430. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0EEW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18041–18041. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0EEX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18042–18042. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BK4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14547–14547. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0EF4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18049–18049. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let I, J be ideals of a Noetherian ring A. Let M be a finite A-module. Let s and d be integers. With T as in (1.0.1) assume*

(1) *A is I-adically complete and has a dualizing complex,*
(2) *if $\mathfrak{p} \in V(I)$ no condition,*
(3) *$cd(A, I) \leq d$,*
(4) *if $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \notin T$ then*

$$depth_{A_\mathfrak{p}}(M_\mathfrak{p}) \geq s \quad or \quad depth_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > d + s$$

for all $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$,
(5) *if $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \notin T$, $V(\mathfrak{p}) \cap V(J) \cap V(I) \neq \emptyset$, and $depth(M_\mathfrak{p}) < s$, then one of the following holds[1]:*
   (a) *$\dim(Supp(M_\mathfrak{p})) < s + 2$[2], or*
   (b) *$\delta(\mathfrak{p}) > d + \delta_{max} - 1$ where δ is a dimension function and δ_{max} is the maximum of δ on $V(J) \cap V(I)$, or*
   (c) *$depth_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > d + s + \delta_{max} - \delta_{min} - 2$ for all $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$.*

Then there exists an ideal $J_0 \subset J$ with $V(J_0) \cap V(I) = V(J) \cap V(I)$ such that for any $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$ the map

$$R\Gamma_{J'}(M) \longrightarrow R\Gamma_J(M)^\wedge$$

induces an isomorphism on cohomology in degrees $\leq s$. Here $^\wedge$ denotes derived I-adic completion.

[1] Our method forces this additional condition. We will return to this (insert future reference).
[2] For example if M satisfies Serre's condition (S_s) on the complement of $V(I) \cup T$.

1. Algebraization of local cohomology, I

Let A be a Noetherian ring and let I and J be two ideals of A. Let M be a finite A-module. In this section we study the cohomology groups of the object

$$R\Gamma_J(M)^\wedge \quad \text{of} \quad D(A)$$

where $^\wedge$ denotes derived I-adic completion. Observe that in Dualizing Complexes, Lemma 0EEX we have shown, if A is complete with respect to I, that there is an isomorphism

$$\operatorname{colim} H^0_Z(M) \longrightarrow H^0(R\Gamma_J(M)^\wedge)$$

where the (directed) colimit is over the closed subsets $Z = V(J')$ with $J' \subset J$ and $V(J') \cap V(I) = V(J) \cap V(I)$. The union of these closed subsets is

$$T = \{\mathfrak{p} \in \operatorname{Spec}(A) : V(\mathfrak{p}) \cap V(I) \subset V(J) \cap V(I)\} \tag{1.0.1}$$

This is a subset of $\operatorname{Spec}(A)$ stable under specialization. The result above becomes the statement that

$$H^0_T(M) \longrightarrow H^0(R\Gamma_J(M)^\wedge)$$

is an isomorphism provided A is complete with respect to I, see Local Cohomology, Lemma 0EF1 and Remark 0EF4. Our method to extend this isomorphism to higher cohomology groups rests on the following lemma.

Lemma 1.1. *Let I, J be ideals of a Noetherian ring A. Let M be a finite A-module. Let $\mathfrak{p} \subset A$ be a prime. Let s and d be integers. Assume*

1. *A has a dualizing complex,*
2. *$\mathfrak{p} \notin V(J) \cap V(I)$,*
3. *$cd(A, I) \leq d$, and*
4. *for all primes $\mathfrak{p}' \subset \mathfrak{p}$ we have $\operatorname{depth}_{A_{\mathfrak{p}'}}(M_{\mathfrak{p}'}) + \dim((A/\mathfrak{p}')_\mathfrak{q}) > d + s$ for all $\mathfrak{q} \in V(\mathfrak{p}') \cap V(J) \cap V(I)$.*

Then there exists an $f \in A$, $f \notin \mathfrak{p}$ which annihilates $H^i(R\Gamma_J(M)^\wedge)$ for $i \leq s$ where $^\wedge$ indicates I-adic completion.

Proof. We will use that $R\Gamma_J = R\Gamma_{V(J)}$ and similarly for $I + J$, see Dualizing Complexes, Lemma 0955. Observe that $R\Gamma_J(M)^\wedge = R\Gamma_I(R\Gamma_J(M))^\wedge = R\Gamma_{I+J}(M)^\wedge$, see Dualizing Complexes, Lemmas 0A6W and 0BJC. Thus we may replace J by $I + J$ and assume $I \subset J$ and $\mathfrak{p} \notin V(J)$. Recall that

$$R\Gamma_J(M)^\wedge = R\operatorname{Hom}_A(R\Gamma_I(A), R\Gamma_J(M))$$

by the description of derived completion in More on Algebra, Lemma 091V combined with the description of local cohomology in Dualizing Complexes, Lemma 0956. Assumption (3) means that $R\Gamma_I(A)$ has nonzero cohomology only in degrees $\leq d$. Using the canonical truncations of $R\Gamma_I(A)$ we find it suffices to show that

$$\operatorname{Ext}^i(N, R\Gamma_J(M))$$

is annihilated by an $f \in A$, $f \notin \mathfrak{p}$ for $i \leq s + d$ and any A-module N. In turn using the canonical truncations for $R\Gamma_J(M)$ we see that it suffices to show $H^i_J(M)$ is annihilated by an $f \in A$, $f \notin \mathfrak{p}$ for $i \leq s + d$. This follows from Local Cohomology, Lemma 0EFE. □

Lemma 1.2. *Let I, J be ideals of a Noetherian ring. Let M be a finite A-module. Let s and d be integers. With T as in (1.0.1) assume*

1. *A has a dualizing complex,*
2. *if $\mathfrak{p} \in V(I)$, then no condition,*
3. *if $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \in T$, then $\dim((A/\mathfrak{p})_\mathfrak{q}) \leq d$ for some $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$,*

(4) *if* $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \notin T$, *then*

$$depth_{A_\mathfrak{p}}(M_\mathfrak{p}) \geq s \quad or \quad depth_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > d + s$$

for all $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$.

Then there exists an ideal $J_0 \subset J$ *with* $V(J_0) \cap V(I) = V(J) \cap V(I)$ *such that for any* $J' \subset J_0$ *with* $V(J') \cap V(I) = V(J) \cap V(I)$ *the map*

$$R\Gamma_{J'}(M) \longrightarrow R\Gamma_{J_0}(M)$$

induces an isomorphism in cohomology in degrees $\leq s$ *and moreover these modules are annihilated by a power of* $J_0 I$.

Proof. Let us consider the set

$$B = \{\mathfrak{p} \notin V(I),\ \mathfrak{p} \in T,\ \text{and } \operatorname{depth}(M_\mathfrak{p}) \leq s\}$$

Choose $J_0 \subset J$ such that $V(J_0)$ is the closure of $B \cup V(J)$.

Claim I: $V(J_0) \cap V(I) = V(J) \cap V(I)$.

Proof of Claim I. The inclusion $\supset$ holds by construction. Let $\mathfrak{p}$ be a minimal prime of $V(J_0)$. If $\mathfrak{p} \in B \cup V(J)$, then either $\mathfrak{p} \in T$ or $\mathfrak{p} \in V(J)$ and in both cases $V(\mathfrak{p}) \cap V(I) \subset V(J) \cap V(I)$ as desired. If $\mathfrak{p} \notin B \cup V(J)$, then $V(\mathfrak{p}) \cap B$ is dense, hence infinite, and we conclude that $\operatorname{depth}(M_\mathfrak{p}) < s$ by Local Cohomology, Lemma 0DWY. In fact, let $V(\mathfrak{p}) \cap B = \{\mathfrak{p}_\lambda\}_{\lambda \in \Lambda}$. Pick $\mathfrak{q}_\lambda \in V(\mathfrak{p}_\lambda) \cap V(J) \cap V(I)$ as in (3). Let $\delta : \operatorname{Spec}(A) \to \mathbf{Z}$ be the dimension function associated to a dualizing complex $\omega_A^\bullet$ for A. Since Λ is infinite and δ is bounded, there exists an infinite subset $\Lambda' \subset \Lambda$ on which $\delta(\mathfrak{q}_\lambda)$ is constant. For $\lambda \in \Lambda'$ we have

$$\operatorname{depth}(M_{\mathfrak{p}_\lambda}) + \delta(\mathfrak{p}_\lambda) - \delta(\mathfrak{q}_\lambda) = \operatorname{depth}(M_{\mathfrak{p}_\lambda}) + \dim((A/\mathfrak{p}_\lambda)_{\mathfrak{q}_\lambda}) \leq d + s$$

by (3) and the definition of B. By the semi-continuity of the function depth $+ \delta$ proved in Duality for Schemes, Lemma 0ECM we conclude that

$$\operatorname{depth}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_{\mathfrak{q}_\lambda}) = \operatorname{depth}(M_\mathfrak{p}) + \delta(\mathfrak{p}) - \delta(\mathfrak{q}_\lambda) \leq d + s$$

Since also $\mathfrak{p} \notin V(I)$ we read off from (4) that $\mathfrak{p} \in T$, i.e., $V(\mathfrak{p}) \cap V(I) \subset V(J) \cap V(I)$. This finishes the proof of Claim I.

Claim II: $H^i_{J_0}(M) \to H^i_J(M)$ is an isomorphism for $i \leq s$ and $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$.

Proof of claim II. Choose $\mathfrak{p} \in V(J')$ not in $V(J_0)$. It suffices to show that $H^i_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ for $i \leq s$, see Local Cohomology, Lemma 0DWV. Observe that $\mathfrak{p} \in T$. Hence since $\mathfrak{p}$ is not in B we see that $\operatorname{depth}(M_\mathfrak{p}) > s$ and the groups vanish by Dualizing Complexes, Lemma 0AVZ.

Claim III. The final statement of the lemma is true.

By Claim II for $i \leq s$ we have

$$H^i_T(M) = H^i_{J_0}(M) = H^i_{J'}(M)$$

for all ideals $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$. See Local Cohomology, Lemma 0EF1. Let us check the hypotheses of Local Cohomology, Proposition 0EFC for the subsets $T \subset T \cup V(I)$, the module M, and the integer s. We have to show that given $\mathfrak{p} \subset \mathfrak{q}$ with $\mathfrak{p} \notin T \cup V(I)$ and $\mathfrak{q} \in T$ we have

$$\operatorname{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > s$$

If $\operatorname{depth}(M_\mathfrak{p}) \geq s$, then this is true because the dimension of $(A/\mathfrak{p})_\mathfrak{q}$ is at least 1. Thus we may assume $\operatorname{depth}(M_\mathfrak{p}) < s$. If $\mathfrak{q} \in V(I)$, then $\mathfrak{q} \in V(J) \cap V(I)$

and the inequality holds by (4). If $\mathfrak{q} \notin V(I)$, then we can use (3) to pick $\mathfrak{q}' \in V(\mathfrak{q}) \cap V(J) \cap V(I)$ with $\dim((A/\mathfrak{q})_{\mathfrak{q}'}) \leq d$. Then assumption (4) gives

$$\text{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_{\mathfrak{q}'}) > s + d$$

Since A is catenary this implies the inequality we want. Applying Local Cohomology, Proposition 0EFC we find $J'' \subset A$ with $V(J'') \subset T \cup V(I)$ such that J'' annihilates $H^i_T(M)$ for $i \leq s$. Then we can write $V(J'') \cup V(J_0) \cup V(I) = V(J'I)$ for some $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$. Replacing J_0 by J' the proof is complete. □

Lemma 1.3. *In Lemma 1.2 if instead of the empty condition (2) we assume*

(2') *if $\mathfrak{p} \in V(I)$, $\mathfrak{p} \notin V(J) \cap V(I)$, then $\text{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > s$ for all $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$,*

then the conditions also imply that $H^i_{J_0}(M)$ is a finite A-module for $i \leq s$.

Proof. Recall that $H^i_{J_0}(M) = H^i_T(M)$, see proof of Lemma 1.2. Thus it suffices to check that for $\mathfrak{p} \notin T$ and $\mathfrak{q} \in T$ with $\mathfrak{p} \subset \mathfrak{q}$ we have $\text{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > s$, see Local Cohomology, Proposition 0EFD. Condition (2') tells us this is true for $\mathfrak{p} \in V(I)$. Since we know $H^i_T(M)$ is annihilated by a power of IJ_0 we know the condition holds if $\mathfrak{p} \notin V(IJ_0)$ by Local Cohomology, Proposition 0EFC. This covers all cases and the proof is complete. □

Lemma 1.4. *If in Lemma 1.2 we additionally assume*

(6) *if $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \in T$, then $\text{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) > s$,*

then $H^i_{J_0}(M) = H^i_J(M) = H^i_{J+I}(M)$ for $i \leq s$ and these modules are annihilated by a power of I.

Proof. Choose $\mathfrak{p} \in V(J)$ or $\mathfrak{p} \in V(J_0)$ but $\mathfrak{p} \notin V(J + I) = V(J_0 + I)$. It suffices to show that $H^i_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ for $i \leq s$, see Local Cohomology, Lemma 0DWV. These groups vanish by condition (6) and Dualizing Complexes, Lemma 0AVZ. The final statement follows from Local Cohomology, Proposition 0EFC. □

Proof. For an ideal $\mathfrak{a} \subset A$ we have $R\Gamma_\mathfrak{a} = R\Gamma_{V(\mathfrak{a})}$, see Dualizing Complexes, Lemma 0955. Next, we observe that

$$R\Gamma_J(M)^\wedge = R\Gamma_I(R\Gamma_J(M))^\wedge = R\Gamma_{I+J}(M)^\wedge = R\Gamma_{I+J'}(M)^\wedge = R\Gamma_I(R\Gamma_{J'}(M))^\wedge = R\Gamma_{J'}(M)^\wedge$$

by Dualizing Complexes, Lemmas 0BJC and 0A6W. This explains how we define the arrow in the statement of the lemma.

We claim that the hypotheses of Lemma 1.2 are implied by our current hypotheses on M. The only thing to verify is hypothesis (3). Thus let $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \in T$. Then $V(\mathfrak{p}) \cap V(I)$ is nonempty as I is contained in the Jacobson radical of A (Algebra, Lemma 05GI). Since $\mathfrak{p} \in T$ we have $V(\mathfrak{p}) \cap V(I) = V(\mathfrak{p}) \cap V(J) \cap V(I)$. Let $\mathfrak{q} \in V(\mathfrak{p}) \cap V(I)$ be the generic point of an irreducible component. We have $\text{cd}(A_\mathfrak{q}, I_\mathfrak{q}) \leq d$ by Local Cohomology, Lemma 0DXB. We have $V(\mathfrak{p}A_\mathfrak{q}) \cap V(I_\mathfrak{q}) = \{\mathfrak{q}A_\mathfrak{q}\}$ by our choice of $\mathfrak{q}$ and we conclude $\dim((A/\mathfrak{p})_\mathfrak{q}) \leq d$ by Local Cohomology, Lemma 0DXF.

Observe that the lemma holds for $s < 0$. This is not a trivial case because it is not a priori clear that $H^i(R\Gamma_J(M)^\wedge)$ is zero for $i < 0$. However, this vanishing was established in Dualizing Complexes, Lemma 0EEW. We will prove the lemma by induction for $s \geq 0$.

The lemma for $s = 0$ follows immediately from the conclusion of Lemma 1.2 and Dualizing Complexes, Lemma 0EEX.

Assume $s > 0$ and the lemma has been shown for smaller values of s. Let $M' \subset M$ be the maximal submodule whose support is contained in $V(I) \cup T$. Then M' is a finite A-module whose support is contained in $V(J') \cup V(I)$ for some ideal $J' \subset J$ with $V(J') \cap V(I) = V(J) \cap V(I)$. We claim that

$$R\Gamma_{J'}(M') \to R\Gamma_J(M')^\wedge$$

is an isomorphism for any choice of J'. Namely, we can choose a short exact sequence $0 \to M_1 \oplus M_2 \to M' \to N \to 0$ with M_1 annihilated by a power of J', with M_2 annihilated by a power of I, and with N annihilated by a power of $I + J'$. Thus it suffices to show that the claim holds for M_1, M_2, and N. In the case of M_1 we see that $R\Gamma_{J'}(M_1) = M_1$ and since M_1 is a finite A-module and I-adically complete we have $M_1^\wedge = M_1$. This proves the claim for M_1 by the initial remarks of the proof. In the case of M_2 we see that $H^i_J(M_2) = H^i_{I+J}(M) = H^i_{I+J'}(M) = H^i_{J'}(M_2)$ are annihilated by a power of I and hence derived complete. Thus the claim in this case also. For N we can use either of the arguments just given. Considering the short exact sequence $0 \to M' \to M \to M/M' \to 0$ we see that it suffices to prove the lemma for M/M'. Thus we may assume $\mathrm{Ass}(M) \cap (V(I) \cup T) = \emptyset$.

Let $\mathfrak{p} \in \mathrm{Ass}(M)$ be such that $V(\mathfrak{p}) \cap V(J) \cap V(I) = \emptyset$. Since I is contained in the Jacobson radical of A this implies that $V(\mathfrak{p}) \cap V(J') = \emptyset$ for any $J' \subset J$ with $V(J') \cap V(I) = V(J) \cap V(I)$. Thus setting $N = H^0_{\mathfrak{p}}(M)$ we see that $R\Gamma_J(N) = R\Gamma_{J'}(N) = 0$ for all $J' \subset J$ with $V(J') \cap V(I) = V(J) \cap V(I)$. In particular $R\Gamma_J(N)^\wedge = 0$. Thus we may replace M by M/N as this changes the structure of M only in primes which do not play a role in conditions (4) or (5). Repeating we may assume that $V(\mathfrak{p}) \cap V(J) \cap V(I) \neq \emptyset$ for all $\mathfrak{p} \in \mathrm{Ass}(M)$.

Assume $\mathrm{Ass}(M) \cap (V(I) \cup T) = \emptyset$ and that $V(\mathfrak{p}) \cap V(J) \cap V(I) \neq \emptyset$ for all $\mathfrak{p} \in \mathrm{Ass}(M)$. Let $\mathfrak{p} \in \mathrm{Ass}(M)$. We want to show that we may apply Lemma 1.1. It is in the verification of this that we will use the supplemental condition (5). Choose $\mathfrak{p}' \subset \mathfrak{p}$ and $\mathfrak{q}' \subset V(\mathfrak{p}) \cap V(J) \cap V(I)$.

(1) If $M_{\mathfrak{p}'} = 0$, then $\mathrm{depth}(M_{\mathfrak{p}'}) = \infty$ and $\mathrm{depth}(M_{\mathfrak{p}'}) + \dim((A/\mathfrak{p}')_{\mathfrak{q}'}) > d + s$.
(2) If $\mathrm{depth}(M_{\mathfrak{p}'}) < s$, then $\mathrm{depth}(M_{\mathfrak{p}'}) + \dim((A/\mathfrak{p}')_{\mathfrak{q}'}) > d + s$ by (4).

In the remaining cases we have $M_{\mathfrak{p}'} \neq 0$ and $\mathrm{depth}(M_{\mathfrak{p}'}) \geq s$. In particular, we see that $\mathfrak{p}'$ is in the support of M and we can choose $\mathfrak{p}'' \subset \mathfrak{p}'$ with $\mathfrak{p}'' \in \mathrm{Ass}(M)$.

(a) Observe that $\dim((A/\mathfrak{p}'')_{\mathfrak{p}'}) \geq \mathrm{depth}(M_{\mathfrak{p}'})$ by Algebra, Lemma 0BK4. If equality holds, then we have

$$\mathrm{depth}(M_{\mathfrak{p}'}) + \dim((A/\mathfrak{p}')_{\mathfrak{q}'}) = \mathrm{depth}(M_{\mathfrak{p}''}) + \dim((A/\mathfrak{p}'')_{\mathfrak{q}'}) > s + d$$

by (4) applied to $\mathfrak{p}''$ and we are done. This means we are only in trouble if $\dim((A/\mathfrak{p}'')_{\mathfrak{p}'}) > \mathrm{depth}(M_{\mathfrak{p}'})$. This implies that $\dim(M_{\mathfrak{p}}) \geq s + 2$. Thus if (5)(a) holds, then this does not occur.

(b) If (5)(b) holds, then we get

$$\mathrm{depth}(M_{\mathfrak{p}'}) + \dim((A/\mathfrak{p}')_{\mathfrak{q}'}) \geq s + \delta(\mathfrak{p}') - \delta(\mathfrak{q}') \geq s + 1 + \delta(\mathfrak{p}) - \delta_{max} > s + d$$

as desired.

(c) If (5)(c) holds, then we get

$$\begin{aligned}
\mathrm{depth}(M_{\mathfrak{p}'}) + \dim((A/\mathfrak{p}')_{\mathfrak{q}'}) &\geq s + \delta(\mathfrak{p}') - \delta(\mathfrak{q}') \\
&\geq s + 1 + \delta(\mathfrak{p}) - \delta(\mathfrak{q}') \\
&= s + 1 + \delta(\mathfrak{p}) - \delta(\mathfrak{q}) + \delta(\mathfrak{q}) - \delta(\mathfrak{q}') \\
&> s + 1 + (s + d + \delta_{max} - \delta_{min} - 2) + \delta(\mathfrak{q}) - \delta(\mathfrak{q}') \\
&\geq 2s + d - 1 \geq s + d
\end{aligned}$$

as desired. Observe that this argument works because we know that a prime $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$ exists.

Now we are ready to do the induction step.

Choose an ideal J_0 as in Lemma 1.2 and an integer $t > 0$ such that $(J_0 I)^t$ annihilates $H^s_J(M)$. The assumptions of Lemma 1.1 are satisfied for every $\mathfrak{p} \in \mathrm{Ass}(M)$ (see previous paragraph). Thus the annihilator $\mathfrak{a} \subset A$ of $H^s(R\Gamma_J(M)^\wedge)$ is not contained in $\mathfrak{p}$ for $\mathfrak{p} \in \mathrm{Ass}(M)$. Thus we can find an $f \in \mathfrak{a}(J_0 I)^t$ not in any associated prime of M which is an annihilator of both $H^s(R\Gamma_J(M)^\wedge)$ and $H^s_J(M)$. Then f is a nonzerodivisor on M and we can consider the short exact sequence

$$0 \to M \xrightarrow{f} M \to M/fM \to 0$$

Our choice of f shows that we obtain

$$\begin{array}{ccccccc} H^{s-1}_{J'}(M) & \longrightarrow & H^{s-1}_{J'}(M/fM) & \longrightarrow & H^s_{J'}(M) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ H^{s-1}(R\Gamma_J(M)^\wedge) & \longrightarrow & H^{s-1}(R\Gamma_J(M/fM)^\wedge) & \longrightarrow & H^s(R\Gamma_J(M)^\wedge) & \longrightarrow & 0 \end{array}$$

for any $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$. Thus if we choose J' such that it works for M and M/fM and $s-1$ (possible by induction hypothesis – see next paragraph), then we conclude that the lemma is true.

To finish the proof we have to show that the module M/fM satisfies the hypotheses (4) and (5) for $s-1$. Thus we let $\mathfrak{p}$ be a prime in the support of M/fM with $\mathrm{depth}((M/fM)_\mathfrak{p}) < s-1$ and with $V(\mathfrak{p}) \cap V(J) \cap V(I)$ nonempty. Then $\dim(M_\mathfrak{p}) = \dim((M/fM)_\mathfrak{p}) + 1$ and $\mathrm{depth}(M_\mathfrak{p}) = \mathrm{depth}((M/fM)_\mathfrak{p}) + 1$. In particular, we know (4) and (5) hold for $\mathfrak{p}$ and M with the original value s. The desired inequalities then follow by inspection. $\square$